# LOW TEMPERATURE THERMODYNAMIC PROPERTIES OF MARTIAN SHERGOTTITES

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**Abstract:** Shergottites are the most abundant type of Martian meteorite found on Earth at ~ 80%. Shergottites are basaltic in composition and similar to terrestrial basalts. Mineralogically, they contain pyroxenes, plagioclase (converted to maskelynite by shock), pigeonite, and augite, while some contain olivine and phyllosilicates. [1,2] We have made low-temperature measurements of the thermodynamic properties of a variety of Martian shergottite meteorites to provide fundamental data for planetary surface modeling and relevant for the Yarkovsky effect, important for their transfer from Mars orbit to Earth. Our data complements previous thermodynamic measurements on regolith made by Mars Viking Landers, and Martian rover missions. [3,4,5,6] We report low-temperature specific heat capacity and thermal conductivity data for eight shergottite (Martian) meteorites: Los Angeles, Northwest Africa [NWA] 7397, NWA 12262, NWA 12269, NWA 12594, NWA 13187, NWA 13366 and NWA 14713, over the range $5 \leq T \leq 300$ K. Bulk density, ρ, was obtained by the Archimedes method after completing the thermodynamic measurements. From these data, we calculate thermal inertia and thermal diffusivity as a function of temperature. Measurements follow procedures described in [7] utilizing a Quantum Design Physical Property Measurement System (QD-PPMS) at Boston College.

## (1) Introduction:

**(a)** Until samples can be returned from Mars by spacecraft, [9] Martian meteorites are the only Mars rock samples currently available for in-depth scientific study and analysis. Sample characterization shown here helps to identify Martian petrographic surface dynamics and aid in future thermodynamic modeling of the Red Planet. [1,8]

**(b)** Heat capacity, thermal conductivity, thermal diffusivity, thermal inertia, bulk density, porosity and thermal expansion are the fundamental thermodynamic material characteristics for meteorites. Such data help to model various aspects of planetary surface properties. In addition, they are essential for understanding rotational/orbital characteristics associated with the transport of these materials from Mars orbit to Earth via the YORP and Yarkovsky effects.

**(c)** Data for $c_p$, κ, $D_T$, and Γ vs. T of the eight shergottite meteorites is displayed in Figs. 1-4. We have interpolated the thermodynamic data values and calculated their respective averages. The variance between the thermodynamic data can be attributed to differences in petrological source material, rock cooling rate, density, porosity and shock history. The interpolated averages (heavy red lines) model the thermodynamic response of shergottite material to a first-order approximation. More shergottites and other Martian meteorite sub-group measurements (nakhlites and chassignites) are necessary to confirm or augment this thermodynamic model.

## (2) Specific Heat Capacity ($c_p$) and Thermal Conductivity (κ) vs. T (5-300 K)

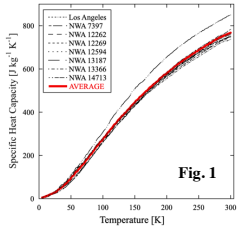
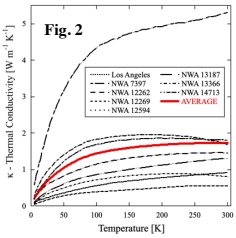



**Fig. 1:** The thin, black, dashed lines represent specific heat capacity data as a function of temperature from the eight individual Martian shergottite samples. The thick red curve represents an average of interpolated specific heat capacity data for all eight samples. Individual data points and error bars are suppressed for clarity and allow comparison of the curve shapes. The averaged values of $c_p$ vary by a factor of 61.1 between 5 to 100 K and by a factor of 2.8 from 100 to 300 K.

**Fig. 2:** The thin black dashed lines represent the κ data as a function of temperature for all Martian shergottite samples. The thick red line represents the average interpolated κ values as a function of T. Individual data points and error bars are suppressed for clarity as in Fig. 1. We note κ vs. T of NWA 7397 is significantly higher than all other samples. Shock analysis by Yoshida [10] on the meteorite NWA 7397 shows that its mineralogy was transformed by heat and pressure during the journey from Mars to Earth to produce these higher κ values.




## (3) Thermal Diffusivity ($D_T$) vs. T (5-300 K)

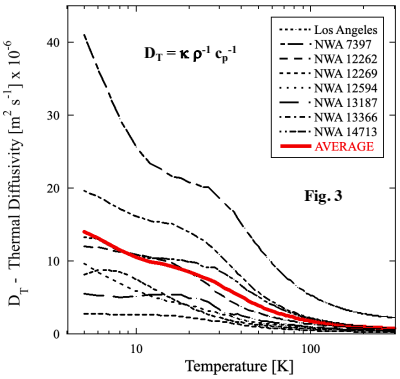



**Fig. 3**: The thin black dashed lines represent the thermal diffusivity as a function of temperature for eight Martian shergottite meteorite samples calculated using $D_{II} = \kappa \rho^{-II} c_{II}^{-II}$. The thick red line represents the average interpolated thermal diffusivity values as a function of temperature for all eight samples. Individual data points and error bars are suppressed for clarity and allow comparison of the curve shapes. The averages values of $D_{II}$ change by a factor of 8.0 from 5 to 100 K and by a factor of 2.3 from 100 K to 300 K.

## (4) Thermal Inertia (Γ) vs. T (5-300 K)

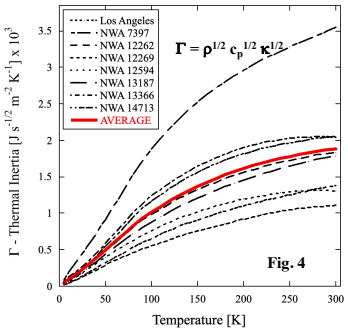
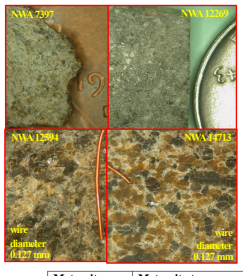



**Fig. 4:** The thin black dashed lines represent the thermal inertia as a function of temperature for eight Martian shergottite meteorite samples calculated using $\Gamma = (\rho\ c_{II}\ \kappa)^{1/2}$. The thick red line represents the average interpolated thermal inertia values as a function of temperature for all eight samples. Individual data points and error bars are suppressed for clarity and allow comparison of the curve shapes. The averaged values of Γ increase by a factor of 21.5 from 5 K to 100 K and by a factor of 1.9 from 100 K to 300 K.

## (5) Martian Shergottite Photos & Density Values




**Photographs:** To the left are four images of Martian shergottites used in these experiments. The upper photographs have a Lincoln cent date mark as a size reference to crystal grain sizes.

The bottom photographs have a small copper wire (diameter = 0.127 mm (0.005 in.) as a size reference. NWA 14713, a lherzolitic shergottite, reveals large olivine (brownish-green) crystals throughout.

| Meteorite | Meteorite type | Bulk density [kg m$^{-3}$] |
|---|---|---|
| Los Angeles | shergottite | 3080.0 +/- 2.5 |
| NWA 7397 | shergottite, lherzolitic (poikilitic) | 3141.8 +/- 1.2 |
| NWA 12262 | shergottite, diabasic | 3022.3 +/- 0.8 |
| NWA 12269 | shergottite, fine-intersertal | 2919.0 +/- 0.6 |
| NWA 12594 | shergottite | 2916.2 +/- 1.4 |
| NWA 13187 | shergottite | 2890.8 +/- 2.0 |
| NWA 13366 | shergottite, peridotitic | 3158.5 +/- 0.7 |
| NWA 14713 | shergottite, lherzolitic | 3114.1 +/- 2.2 |
| AVERAGE | | 3030.3 +/- 1.4 |

## (6) Conclusions/Analysis:

**(a)** This is the first systematic, low-temperature thermodynamic study of Martian rocks; a first for any Martian shergottite meteorites at T<100 K.

**(b)** Thermal conductivity and thermal inertia values are much higher than those values measured for regolith by Viking landers and Mars rovers. [3,4,5,6] Differences stem from the high regolith porosity, which suppresses both the thermal conductivity and density of the material.

**(c)** The "average" shergottite data for $c_p$, κ, $D_T$, and Γ vs. T (5-300 K) can assist in modeling the thermal behavior of the Martian crust and understanding how Martian rocks influence the YORP and Yarkovsky effects.

**(d)** The values of $c_p$ and κ for NWA 7397 are noticeably higher than the other data. These values are likely influenced by the shock state this meteorite developed while being ejected from Mars and upon entering Earth's atmosphere. [10]

**References:** [1] McSween Jr. H. Y. (2015) *Amer. Miner.* 100: 2380–2395. [2] McSween, H. Y., and Treiman, A. H. (1998) *Rev. in Mineralogy*, 36: 6-1 to 6-53. [3] Edwards C. S. et al. (2018) J. of Geophys. Res. Planets 123:1307–1326. [4] Grott et al. (2021) *J. Geophys. Res. Planets* 126: e2021JE006861. [5] Piqueux S. et al. (2021) *J. Geophys. Res. Planets* 126: e2021JE006859. [6] Singh D. et al. (2022) Earth and Planet. Phys. 6:18–27. [7] Opeil C. et al. (2020) *MAPS* 55: E1-E20. [8] Treiman A. H. and Filiberto J. (2015) *MAPS* 50: 632– 648. [9] Udry A. et al. *J. Geophys. Res. Planets* 125: e2020JE006523. [10] Yoshida M. et al. (2021) *MAPS* 56: 1729- 1743.

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